The University of Sydney

MOOC Introduction to Calculus

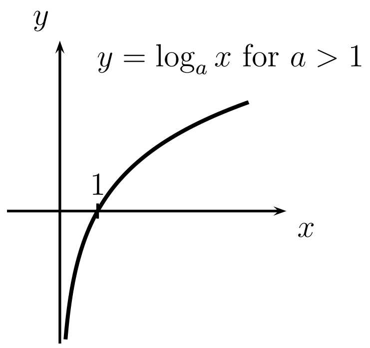
Notes for 'The logarithmic function'

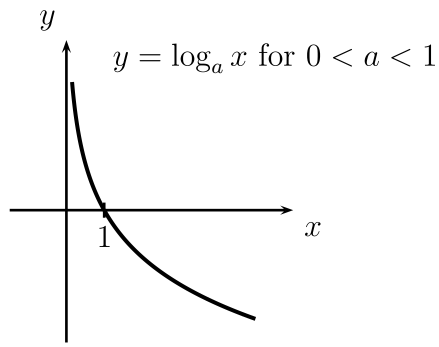
Important Ideas and Useful Facts:

(i) Logarithmic functions: Suppose a is positive and $a \neq 1$. The inverse of the function $y = a^x$ is called the *logarithmic function to the base* a and denoted by

$$y \;=\; \log_a(x)\ .$$

The graph of the logarithm function is obtained by reflecting the graph of $y = a^x$ in the line $y = x$. Its domain is $(0, \infty)$, its range is $\mathbb{R}$ and the y-axis is a vertical asymptote.





In the special case that $a = e$, where e is Euler's number, we write

$$y \;=\; \log_e x \;=\; \ln x\ ,$$

and $\ln x$ is called the *natural logarithm* of x.

(ii) Logs and exponentials undo each other: For all x, all positive A and all positive $a \neq 1$,

$$\log_a(a^x) \;=\; x\,, \qquad \ln e^x \;=\; x\,, \qquad a^{\log_a A} \;=\; A\,, \qquad e^{\ln A} \;=\; A\,.$$

(iii) Logarithmic laws: For all positive x and y, for all positive $a \neq 1$, and for all real k,

(a) $\log_a(xy) = \log_a x + \log_a y\,,\quad \log_a(x/y) = \log_a x - \log_a y\,,\quad \log_a(x^k) = k\log_a x\,.$

(b) $\ln(xy) = \ln x + \ln y\,,\quad \ln(x/y) = \ln x - \ln y\,,\quad \ln(x^k) = k\ln x\,.$

(c) $a^x = e^{x\ln a}\,,\quad \log_a x = \dfrac{\ln x}{\ln a}\ .$

Examples and derivations:

1. Since $2^5 = 32$ and $3^7 = 2,187$ we have

$$\log_2(32) \;=\; \log_2(2^5) \;=\; 5 \qquad \text{and} \qquad \log_3(2,187) \;=\; \log_3(3^7) \;=\; 7\ .$$

2. Using properties of logarithms, we quickly simplify the following expression:

$$\log_{10}\sqrt{1,000} \;=\; \log_{10}\left(1,000^{1/2}\right) \;=\; \frac{1}{2}\log_{10}(1,000) \;=\; \frac{1}{2}\log_{10}(10^3) \;=\; \frac{3}{2}\ .$$

3. Let's solve the following equation for x:

$$7^x \;=\; 25\;.$$

By taking natural logarithms of both sides, we get

$$x\ln 7 \;=\; \ln(7^x) \;=\; \ln(25)\;,$$

yielding the solution

$$x \;=\; \frac{\ln(25)}{\ln 7}\;.$$

4. Let's solve the following equation for x:

$$9^x \;=\; e^{2x-3}\;.$$

By taking natural logarithms of both sides, we get

$$x\ln 9 \;=\; \ln(e^{2x-3}) \;=\; 2x-3$$

so that

$$x(\ln 9 - 2) \;=\; -3\;,$$

yielding the solution

$$x \;=\; \frac{-3}{\ln 9 - 2} \;=\; \frac{3}{2-\ln 9}\;.$$

5. We explain how the logarithm laws follow from the exponential laws. Suppose that x and y are positive, that a is positive but $a \neq 1$, and that $k \in \mathbb{R}$.

Observe first that $x = a^b$ and $y = a^c$ for some real numbers b and c, namely

$$b \;=\; \log_a x \qquad \text{and} \qquad c \;=\; \log_a y\;.$$

Then, because $\log_a$ is the inverse function for the exponential function to the base a, and using an exponential law, we have

$$\log_a(xy) \;=\; \log_a(a^b a^c) \;=\; \log_a(a^{b+c}) \;=\; b+c \;=\; \log_a x + \log_a y\;,$$

verifying the first logarithm law in part (a) of (iii). By another exponential law we have

$$\log_a\left(\frac{x}{y}\right) \;=\; \log_a\left(\frac{a^b}{a^c}\right) \;=\; \log_a(a^{b-c}) \;=\; b-c \;=\; \log_a x - \log_a y\;,$$

verifying the second logarithm law in part (a) of (iii). By yet another exponential law we have

$$\log_a(x^k) \;=\; \log_a\left((a^b)^k\right) \;=\; \log_a(a^{bk}) \;=\; bk \;=\; kb \;=\; k\log_a x\;,$$

verifying the third logarithm law in part (a) of (iii). The laws in part (b) of (iii) are immediate, since $\ln = \log_e$. Observe, again by an exponential law, that

$$a^x \;=\; \left(e^{\ln a}\right)^x \;=\; e^{(\ln a)x} \;=\; e^{x\ln a}\;,$$

verifying the first law in part (c) of (iii). By the third law in part (b) of (iii),

$$\ln x \;=\; \ln(a^b) \;=\; b\ln a \;=\; (\log_a x)(\ln a)\;,$$

so that

$$\log_a x \;=\; \frac{\ln x}{\ln a}\;,$$

since $\ln a \neq 0$, as $a \neq 1$, verifying the second law in part (c) of (iii).